Renaldo Miya

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EDUCATION

University of Waterloo **Expected Graduation: April 2029**
BASc, Honours Mechanical Engineering

WORK EXPERIENCE

Design Engineering Intern **September 2026 – December 2026**
Amphenol Markham, ON

- Redesigned an RJ45 housing snap-fit latch to exceed a competitor's retention force, reworking beam geometry to cut peak root stress 25%. Validated hand calcs against SolidWorks Simulation FEA and released to production
- Designed an IEC 60999-1 test fixture to measure plug contact resistance under dynamic motion as a lower-cost alternative to a $2K commercial rig; secured management approval and coordinated fabrication
- Released GD&T detail and manufacturing drawings for 10+ connector and receptacle part numbers
- Built SolidWorks drawing templates that cut update time ~30% across 5 product lines

Mechanical Engineering Assistant **January 2026 – April 2026**
TRIUMF | Canada's national particle accelerator laboratory Vancouver, BC

- Owned the design of a wall-mounted aluminium support frame for ARIEL's radioactive-gas pumping manifold, from load and radiation-area requirements to fabrication-ready drawings
- Iterated clamp placement in ANSYS to cut peak manifold deflection 83% (6.5 → 1.13 mm) and peak tubing stress 54% (107 → 49.6 MPa), raising the yield safety factor from 1.6 to 3.4; validated against hand calcs
- Placed Unistrut posts at the FEA clamp locations and sized the frame so the clamp points barely deflect, reducing part count 25% from the first design to shorten install time and dose exposure
- Released ASME Y14.5 GD&T drawings and managed BOMs and revisions in SolidWorks PDM; modelled beamline shielding components and revised them through design reviews with senior engineers

Mechanical Engineering Co-op **May 2024 – August 2024**
BC Hydro | Site C Clean Energy Project Fort St John, BC

- Reviewed and redlined P&IDs and mechanical layouts against applicable codes and standards, flagging constructability issues and issuing modifications adopted into the build
- Conducted on-site field inspections to verify installation accuracy and code compliance, documenting discrepancies and driving corrective design changes that reduced rework

PROJECTS & DESIGN TEAMS

Force-Feedback Robotic Gripper | SolidWorks, Arduino Uno (C++), 3D Printing **Fall 2026 – Present**

- Designed a two-finger, servo-actuated rack and pinion gripper in SolidWorks for FDM printing, using a PLA frame with flexible TPU finger pads for better grip
- Integrated FSR 402 force sensors with an Arduino Uno to run a closed-loop proportional controller that adjusts servo position to hold a target grip force, handling fragile and rigid objects without crushing them
- Calculated gear module and tooth profiles to minimize backlash within the rack and pinion mechanism, ensuring precise positioning of the 3D-printed PLA/TPU fingers

UW Formula Electric | Mechanical Design & Composites Member **2025**

- Designed a 3D-printed enclosure for the vehicle's power distribution relays in SolidWorks, cutting mass 20% from the first iteration and reaching a flush fit with fixed chassis mounts and connectors in 3 printed iterations
- Built composite parts by hand, moulding the undertray and laying up carbon fibre plating, and machined components on shop equipment

TECHNICAL SKILLS

CAD & Modelling | SolidWorks, AutoCAD, DraftSight
Analysis & Simulation | ANSYS (Mechanical, Fluent), SolidWorks Simulation, MATLAB
Prototyping & Fabrication| 3D Printing, Silicone Moulding, CNC Machining, Mill, Lathe, Drill Press, Composite Layup
Design & Documentation | GD&T (ASME Y14.5), DFM/DFA, Fixture Design, PDM, Windchill
Programming | C++, C, Python, HTML/CSS